

■ Centralized Power Distribution System

PCB & Hardware Reference Document
IIT Hyderabad — Department of Electrical Engineering

Document	PCB & Hardware Reference
Project	Centralized Power Distribution System (CPDS)
Institution	IIT Hyderabad — EE Laboratory
PCB Tool	KiCad 7.x
Board	2-layer custom PCB — XIAO ESP32-C3 + ULN2003 + 3× Relay
Boards Total	10 (one per lab table, T-01 ... T-10)
Revision	v3.0 Academic Year 2025–26

This document is the complete hardware reference for the CPDS project. It covers the system overview, schematic design, PCB layout specifications, assembled board photos, GPIO pin assignments, power architecture, full Bill of Materials, component footprints, Gerber file list, and assembly guidelines. For firmware and web portal documentation, refer to the project README.

1. System Overview

Each of the 10 lab tables hosts one custom PCB. The board is built around a **Seeed Studio XIAO ESP32-C3** which polls Firebase Realtime Database every 2 seconds for its *isOn* flag. When the portal sets *isOn* = *true* for a table, the ESP32 drives three **5V relay modules** through a **ULN2003APG Darlington array**, closing the socket contacts. A **Hi-Link HLK-PM01** isolated SMPS converts 230V AC mains to the 5V operating rail. A **18650 Li-Ion cell** charged by **TP4056** and boosted by **MT3608** provides automatic backup if mains is lost. The **AO3401A P-MOSFET** switches between the mains and battery rails without firmware intervention. A **bi-color LED** shows power source status: Green = mains, Red = battery/error.

■ *system_overview.jpg* — Wide shot of full table setup with PCB, relays, and socket rail
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Simplified Block Diagram

AC Mains 230V	→	HLK-PM01 Isolated SMPS	→	5V DC Rail
18650 Li-Ion	→	TP4056 → cell → MT3608	→	5V DC Rail (backup)
5V DC Rail	→	XIAO ESP32-C3 + ULN2003APG	→	3× Relay Modules → Sockets
XIAO ESP32-C3	→	Wi-Fi → Firebase RTDB		Web Portal ↔ ESP32

2. Schematic

The schematic is designed in KiCad 7. The circuit is organized into four functional blocks: (A) AC mains input, varistor protection, and HLK-PM01 isolated SMPS; (B) Battery management — TP4056 charger, 18650 cell, MT3608 boost, AO3401A MOSFET switch; (C) ESP32-C3 microcontroller, mains-sense voltage divider, emergency button, LED; (D) ULN2003APG relay driver and three 5V relay module outputs.

■ **schematic.png** — Full KiCad schematic export (File → Plot → PNG in KiCad)

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Key Design Decisions

HLK-PM01 isolated SMPS

Provides 3kV galvanic isolation between 230V AC mains and the 5V DC side. No custom transformer design needed — module is certified.

ULN2003APG Darlington array

Open-collector outputs can sink up to 500mA per channel at 50V. Drives relay coils directly from ESP32 3.3V logic without additional transistors.

AO3401A P-channel MOSFET

Automatic battery failover without firmware involvement. When HLK-PM01 output drops, the MOSFET gate control switches the battery path ON. Driven via NPN transistor level-shift because ESP32 GPIO is 3.3V but MOSFET needs 5V gate drive.

ADC mains sense (voltage divider 120kΩ / 15kΩ)

Scales the 5V mains rail to ~0.55V for the 12-bit ADC. Threshold set to ADC reading 500. 100μF + 22μF capacitors filter switching noise. 5-reading hysteresis prevents false triggers.

SS34 Schottky diodes on relay flyback

Placed on-board (not relying on relay module's internal diode) for reliable coil back-EMF suppression.

PCB slot in Edge.Cuts

Physical air gap cut in the PCB between AC and DC sections. Combined with 3mm trace clearance, this meets IPC-2221 creepage requirements.

3. PCB Layout

■ *pcb_front.png* — KiCad 3D Viewer, top view (front copper F.Cu)
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■ *pcb_back.png* — KiCad 3D Viewer, flipped (back copper B.Cu)
Add image here — see media/ folder

■ *pcb_3d_top.png* — KiCad 3D Viewer raytracing render, top-down
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PCB Specifications

Parameter	Value
Design Tool	KiCad 7.x
Layers	2 (F.Cu signal/power + B.Cu ground plane)
Mains trace width	1.5 mm minimum
Signal trace width	0.3 mm minimum
Mains clearance	3.0 mm minimum to DC traces + PCB slot
DC clearance	0.2 mm minimum
Surface finish	HASL (Hot Air Solder Leveling)
Solder mask	Green, both sides
Silkscreen	White, both sides — reference designators + MAINS DANGER warning

Parameter	Value
Board thickness	1.6 mm
Copper weight	1 oz (35 µm)

4. Assembled Board

■ **pcb_assembled.jpg** — *Real photo of soldered, working PCB (bird's-eye, good lighting)*

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■ **table_assembly.jpg** — *Control box or PCB mounted at table with wiring visible*

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5. Pin Map — XIAO ESP32-C3

The XIAO ESP32-C3 has 11 usable GPIO pins. The table below lists all assignments for this project. All relay outputs are active-HIGH (HIGH = relay energised = socket ON). The emergency button uses INPUT_PULLUP — active LOW when pressed.

Pin Label	GPIO #	Direction	Function	Notes
D0	2	INPUT (ADC)	Mains sense	Voltage divider 120kΩ/15kΩ → ~0.55V at 5V. Threshold: ADC 500
D1	3	INPUT_PULLUP	Emergency button	NO contact to GND. Active LOW. 50ms debounce.
D2	4	OUTPUT	Relay 3 drive	HIGH = relay 3 coil ON via ULN2003 I3 pin
D3	5	OUTPUT	Relay 2 drive	HIGH = relay 2 coil ON via ULN2003 I2 pin
D4	6	OUTPUT	Relay 1 drive	HIGH = relay 1 coil ON via ULN2003 I1 pin
D5	7	OUTPUT	Green LED	HIGH = LED ON. Mains present, no error condition
D10	10	OUTPUT	Red LED	HIGH = LED ON. Battery mode, error, or boot phase

Firmware Timing Constants (lab_controller_v3.ino)

#define Constant	Value	Purpose
MAINS_THRESHOLD	500	ADC reading below this = mains lost (12-bit ADC, 0–4095 range)
DEBOUNCE_MS	50 ms	Emergency button software debounce window
FIREBASE_POLL_MS	2000 ms	How often ESP reads tables/T-XX/isOn from Firebase
HEARTBEAT_INTERVAL_MS	10000 ms	How often ESP writes power/T-XX/source and lastSeen

6. Bill of Materials (BOM)

Quantities shown are for the full 10-table lab deployment (not per-board). Multiply relay quantity by 3 per board = 30 total relay modules for 10 boards. All other components: quantity shown is for all 10 boards combined.

#	Component	Description	MPN	Qty
1	Seeed Studio XIAO ESP32-C3	Wi-Fi MCU, USB-C, castellated	Generic	10
2	5V 1-Ch Relay Module	SRD-05VDC-SL-C coil, 10A contacts	SRD-05VDC-SL-C	30
3	ULN2003APG (Toshiba)	50V/500mA 7-ch Darlington relay driver	ULN2003APG	10
4	Momentary Push Switch	NO contact, active-low emergency button	TSD001B07018A	10
5	Hi-Link HLK-PM01	230VAC → 5V/3W isolated SMPS	HLK-PM01	10
6	18650 Li-Ion Battery	3.7V, 2500mAh backup cell	INR18650-25P	10
7	18650 Battery Holder SMD	Single cell SMD/SMT holder	1042P	10
8	TP4056 Charging Module	Adjustable 1A Li-Ion charger	TP4056	10
9	MT3608 Boost Converter	2A max DC-DC step-up to 5V	MT3608	10
10	Bi-Color LED 5mm	Red/Green dual-color indicator	XL-A524SURSYPGW	20
11	6-Pin DPDT Self-Lock Switch	Latching power switch	—	10
12	AO3401A MOSFET	P-channel SOT-23-3, battery path switch	A03401A	10
13	3-Pin Screw Terminal 3.5mm	Mains & load connections	XY302V-3.5-3P	30
14	100µF / 35V Capacitor DIP	Electrolytic, 5V rail bulk cap	ERS1CM101D110T	20
15	22µF Capacitor DIP	Electrolytic, bypass / ADC filter	ERS1KM220E120T	20
16	SS34 Schottky Diode	3A/40V, relay flyback + path blocking	—	30
17	22µH Inductor SMD	MT3608 boost converter inductor	ASPI-0630LR-220M-T15	10
18	1.2kΩ Resistor THT	LED / GPIO current limiting	MF25 1K2	10
19	330Ω Resistor THT	LED series resistor	—	20
20	120kΩ Resistor THT	Voltage divider upper leg + gate pull-up	—	10
21	15kΩ Resistor THT	Voltage divider lower leg	—	10
22	Varistor D15.5mm P7.5mm	MOV surge protection on AC input	561KD10J	10

7. Component Footprints

Standard footprints are from KiCad's built-in libraries. Custom footprints are in the **kicad/mosfet.pretty/** folder — stored as a relative path so the PCB opens correctly on any machine as long as the repo folder structure is preserved.

Component	KiCad Footprint String	Source
XIAO ESP32-C3	SeeedStudio_XIAO_ESP32C3	Custom
Relay Module	Relay_THT:HF41F	Standard
ULN2003APG	Package_DIP:DIP-16_W7.62mm	Standard
Momentary Push Switch	Button_Switch_THT:SW_PUSH_6mm	Standard
HLK-PM01	Converter_ACDC:HLK-PM01	Standard
18650 Battery Holder	BatteryHolder_THT:BatteryHolder_Keystone_1042P	Standard
TP4056 Module	Module_TP4056	Custom
MT3608 Boost Module	Module_MT3608	Custom
AO3401A MOSFET	mosfet.pretty:SOT23-3	Custom — kicad/mosfet.pretty/
Bi-Color LED 5mm	LED_THT:LED_D5.0mm_2pin	Standard
3-Pin Screw Terminal	TerminalBlock_Phoenix:MKDS_3-3.5	Standard
100µF Capacitor DIP	Capacitor_THT:CP_Radial_D8.0mm_P3.50mm	Standard
22µF Capacitor DIP	Capacitor_THT:CP_Radial_D6.3mm_P2.50mm	Standard
SS34 Schottky Diode	Diode_THT:D_D0-201AD	Standard
22µH Inductor SMD	Inductor_SMD:L_Bourns_SRR1260	Standard
All THT Resistors	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm	Standard
Varistor D15.5mm P7.5mm	Varistor_THT:RV_D15.5mm_P7.5mm	Standard

8. Gerber Files (kicad/lab_output/)

All Gerber files are pre-exported and stored in **kicad/lab_output/**. To order PCBs: ZIP the entire **lab_output/** folder and upload to JLCPCB, PCBWay, or any manufacturer.

File Name	Layer / Contents
lab-F_Cu.gbr	Front copper — signal traces, relay drive lines, power tracks
lab-B_Cu.gbr	Back copper — full ground plane
lab-F_Mask.gbr	Front solder mask — defines exposed pads on top side
lab-B_Mask.gbr	Back solder mask — defines exposed pads on bottom
lab-F_Silkscreen.gbr	Front silkscreen — component references, polarity marks, MAINS DANGER label
lab-B_Silkscreen.gbr	Back silkscreen
lab-F_Paste.gbr	Front solder paste — for SMD reflow (AO3401A, inductor)
lab-B_Paste.gbr	Back solder paste
lab-Edge_Cuts.gbr	Board outline + PCB isolation slot between AC and DC sections
lab-PTH.dr1	Plated through-hole drill file — component leads, via holes
lab-NPTH.dr1	Non-plated through-hole drill file — mounting holes
lab-job.gbrjob	Gerber job metadata — layer list and file associations

9. Assembly Guidelines

■ **WARNING** — This board handles 230V AC mains voltage. Treat all AC-side connections with extreme caution.

Step 1. SMD components first

Solder AO3401A (SOT-23-3) and 22μH SMD inductor before through-hole components. Use soldering iron at 330°C with fine tip.

Step 2. Through-hole resistors and diodes

Solder all THT resistors and SS34 Schottky diodes. Observe diode polarity — cathode band toward anode side of symbol.

Step 3. Electrolytic capacitors

Solder 100μF and 22μF capacitors. Polarity is critical — reversed electrolytics can rupture under power.

Step 4. Screw terminals, LED, and switch

Solder 3-pin screw terminals, bi-color LED (observe polarity), and NO emergency push button.

Step 5. Module pin headers

Solder pin headers for relay modules, TP4056, and MT3608. Do NOT solder modules directly — headers allow replacement.

Step 6. XIAO ESP32-C3

Solder XIAO on castellated pads or pin headers. Ensure all pads are reflowed and bridged correctly.

Step 7. HLK-PM01 last

Mount the isolated SMPS module last. Verify AC and DC sections are fully separated before powering on.

Step 8. 18650 cell

Insert 18650 battery after full board inspection. Observe polarity on holder.

Step 9. Bench test before mains connection

Power from a 5V DC bench supply first. Verify relay clicks on command, LED responds, ESP32 boots correctly via USB serial monitor. Only after this test connect the HLK-PM01 to mains.

10. Images Required for This Document

The placeholder boxes throughout this document correspond to the following image files. Add them to the correct folders in the repository — the README and this PDF will reference them automatically. See IMAGES_GUIDE.md in the repo for step-by-step instructions.

File Name	Folder	How to Get It	Section
system_overview.jpg	media/hardware/	Real photo of full table	Cover + README
schematic.png	media/hardware/	KiCad File → Plot → PNG	Section 2
pcb_front.png	media/hardware/	KiCad 3D Viewer → top	Section 3
pcb_back.png	media/hardware/	KiCad 3D Viewer → flipped	Section 3
pcb_3d_top.png	media/hardware/	KiCad 3D Viewer raytracing	Section 3
pcb_assembled.jpg	media/hardware/	Real photo of soldered PCB	Section 4
table_assembly.jpg	media/hardware/	Real photo, control box	Section 4
01_home_page.png	media/screenshots/	Browser screenshot	README §10
02_student_roll_entry.png	media/screenshots/	Browser screenshot	README §10
03_student_table_select.png	media/screenshots/	Browser screenshot	README §10
04_student_confirmation.png	media/screenshots/	Browser screenshot	README §10
05_staff_login.png	media/screenshots/	Browser screenshot	README §10
06_staff_dashboard.png	media/screenshots/	Browser screenshot	README §10
demo_video.mp4	media/	Screen + hardware recording	README §11

For full system documentation — software architecture, execution flow, Firebase schema, setup instructions, and portal walkthrough — refer to **README.md** at the repository root.